Homotopical Encrypt: Hidden Hodge Operators and the Harmonic Cochain Problem

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Abstract

We introduce the Harmonic Cochain Problem (HCP), the problem of recovering the harmonic cochain $h \in \ker(\Delta)$ selected by a hidden Hodge–Laplacian Δ from a public homotopy class, without determining Δ or $\ker(\Delta)$. A hidden representative determines restriction data, adjacent coboundaries δ_0, δ_1, the degree-one Hodge–Laplacian $\Delta_1 = \delta_0\delta_0^* + \delta_1^*\delta_1$, and the Hodge projection $\pi_H : C^1 \to \ker(\Delta_1)$ used for recovery; the public class records only a classifier shadow, while the observed cochain has the form $b = h + e$, where $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$ is Hodge-structured noise. For every explicitly generated lattice $\Lambda_B = \mathrm{im}(B)$, there is a linear operator Δ with $\ker(\Delta) = \Lambda_B$; hence a presented-operator closest-harmonic solver returns a closest vector for the CVP instance (B, t). Conversely, a correct known-lattice CVP oracle does not uniformly solve public-data HCP when publicly equivalent hidden operators may disagree on harmonicity. We isolate the semantic and axiomatic surfaces needed for this statement, including fibrational identity data, homotopy classes, exact invariants, spectral/Tsirelson admissibility, and Heyting-style oracle judgments. We then define the public artifact stack of cochain commitment payloads, commitment axioms, proof-carrying artifacts, and lossy projections, and construct the finite projection endpoint as a Reed–Solomon mutual-correlated-agreement statement. The definitions, theorem boundaries, and Lean correspondence are formalized and explicitly indexed in the appendix ledger.

Contents

1 Introduction **3**
1.1 Motivation 3
1.2 Contributions 3

2 The Harmonic Cochain Problem **4**
2.1 Public And Hidden Data 6

2.2 Recovery Target 8
2.3 Computational Model 8

3 CVP Comparison And Kernel Ambiguity 9
3.1 Known-Lattice CVP Surface 9
3.2 Reduction Direction 10
3.3 Oracle Insufficiency 12
3.4 Structural Comparison 14

4 Axioms, Semantics, And Public Data 14
4.1 Fibrational Semantics 14
4.2 Fibrational Identity Objects 15
4.3 Homotopy Classes 16
4.4 Axiom And Interface Surfaces 16
4.5 Exact Structural Invariants 18
4.6 Spectral-Gap Admissibility 19
4.7 Heyting And Oracle Judgment Surface 24

5 Public Artifact Stack And Lossy Projection 25
5.1 Public, Artifact, And Private Layers 25
5.2 The Artifact Stack 26
5.3 Cochain Commitments 26
5.4 Commitment Map 26
5.5 Commitment And Artifact Invariants 28
5.6 Sheaf Storage And Witness Material 29
5.7 Lossy Projection 29
5.8 Preserved And Lost Structure 29

6 Reed–Solomon Projection And MCA 30
6.1 Projection Chain 30
6.2 Projection-Stage Gates 31
6.3 Canonical Reed–Solomon Endpoint 32
6.4 Mutual Correlated Agreement 32
6.5 Corollary 33

7 Future Directions 34
7.1 Protocol Boundary 34
7.2 Complexity-Theoretic Refinements 34

A Lean/Paper Correspondence **35**
A.1 Definitions And Interfaces 35
A.2 Theorems And Status 36

B Complete Lean Coverage Ledger **38**
B.1 Build And Aggregate Files 38
B.2 HCP, CVP, And Fibrational Algebra 38
B.3 Foundational, Sheaf, Spectral, And Judgment Axioms 39
B.4 Laplacian Geometry And Search Spaces 40
B.5 Commitments, Public Artifacts, And TCB Boundary 41
B.6 Projection Chain, Proximity Gates, And RS/MCA Endpoint 41
B.7 Subsequent Protocol And Implementation Boundary 43

1 Introduction

1.1 Motivation

Homotopical Encrypt is organized around the following security shape:

$$\begin{array}{c}\text{public handles}\\ \text{and verifier-facing artifacts}\end{array} \quad \not\Rightarrow \quad \text{the operative Hodge–Laplacian kernel.}$$

The private recovery object is the harmonic cochain selected by a hidden Hodge operator, instead of a scalar or a vector secret in a published lattice or algebraic object. Public verification therefore needs finite artifacts, while security must not collapse to recovering the private representative or the hidden operator.

1.2 Contributions

This paper develops the following contributions.

(i) The Harmonic Cochain Problem (HCP): recovery of the harmonic cochain selected by a hidden Hodge operator.

(ii) A Hodge-structured noise model with cochain decomposition $b = h + e$, where $h \in \ker(\Delta)$ and $e \in \mathrm{im}(\delta) \oplus \mathrm{im}(\delta^*)$.

(iii) A CVP comparison in which an explicit lattice is realized as a kernel $\ker(\Delta)$, a harmonic solver yields a lattice vector, and the closest-harmonic contract recovers the closest-vector predicate.

(iv) A hidden-operator ambiguity theorem: publicly equivalent Hodge operators may disagree on harmonicity, so a correct explicit-lattice CVP oracle does not uniformly solve public-data HCP.

(v) A Lean formalization of the definitions, reductions, projection chain, gates, and theorem boundaries.

(vi) A projection chain through Plucker shadows and cochain commitments to a canonical Reed–Solomon endpoint with an MCA bad-challenge bound.

2 The Harmonic Cochain Problem

Definition 2.1 (Harmonic Cochain Problem)**.** *The* Harmonic Cochain Problem *(HCP) asks, given public cochain/class-level data associated to a Hodge system, to recover the harmonic cochain determined by the operative Laplacian kernel.*

In the Homotopical Encrypt public-sample model, the visible cochain has the form

$$b = h + e, \qquad h \in \ker(\Delta_1), \qquad e \in \operatorname{im}(\delta_0) \oplus \operatorname{im}(\delta_1^*).$$

Here h is the harmonic cochain, equivalently the global-section component, and e is Hodge-structured noise. The hidden-operator condition is that the public data does not determine a unique Δ_1 or $\ker(\Delta_1)$. The algebraic recovery chain is

$$a \longmapsto \{R_{v\to e}\} \longmapsto (\delta_0, \delta_1) \longmapsto \Delta_1 \longmapsto \pi_H : C^1 \to \ker(\Delta_1),$$

where a is a hidden representative, $R_{v\to e}$ are the restriction maps determining the adjacent coboundaries

$$\delta_0 : C^0 \to C^1, \qquad \delta_1 : C^1 \to C^2,$$

and

$$\Delta_1 = \delta_0\delta_0^* + \delta_1^*\delta_1$$

is the induced degree-one Hodge–Laplacian. The map π_H is the orthogonal projection onto $\ker(\Delta_1)$. When the degree is fixed, we write Δ for this operative Laplacian. The public observation may bind to this chain, but does not expose the chain.

Definition 2.2 (Matrix harmonicity)**.** *For a hidden linear Hodge operator Δ, a cochain $h \in R^N$ is Δ-harmonic when*

$$\Delta h = 0.$$

Thus the harmonic kernel is $\ker(\Delta)$*. In the projection model this agrees with the condition $h \in W$ when $\Delta = \Delta_W = I - P_W$.*

Definition 2.3 (Projection Laplacian, harmonic projection, and residual)**.** *Let $V = R^n$ be equipped with its standard inner product, and let $W \subset V$ be a linear subspace. Write*

$$P_W : V \to W \subset V \qquad \textit{and} \qquad P_{W^\perp} : V \to W^\perp \subset V$$

for the orthogonal projections. The projection-model Laplacian attached to W is

$$\Delta_W = I - P_W.$$

For $b \in V$, define the harmonic projection, residual vector, and residual score by

$$\pi_W(b) = P_W b, \qquad r_W(b) = \Delta_W b, \qquad \rho_W(b) = \|r_W(b)\|.$$

We say that b is W-harmonic when $r_W(b) = 0$.

Proposition 2.4 (Projection harmonicity equivalences)**.** *For every $b \in V$,*

$$r_W(b) = P_{W^\perp} b, \qquad \rho_W(b) = \|P_{W^\perp} b\|, \qquad \rho_W(b) \geq 0.$$

Moreover

$$\pi_W(b) \in W, \qquad r_W(\pi_W(b)) = 0,$$

and the following conditions are equivalent:

$$r_W(b) = 0 \quad \iff \quad \rho_W(b) = 0 \quad \iff \quad b \in W.$$

Proof. Since $V = W \oplus W^\perp$ orthogonally, every $b \in V$ decomposes uniquely as

$$b = P_W b + P_{W^\perp} b.$$

Hence

$$(I - P_W)b = b - P_W b = P_{W^\perp} b,$$

giving $r_W(b) = P_{W^\perp} b$ and the displayed formula for $\rho_W(b)$. Nonnegativity is the nonnegativity of the norm. The projection $P_W b$ lies in W, so its orthogonal component is zero; therefore $r_W(\pi_W(b)) = 0$. Finally, $r_W(b) = 0$ iff $P_{W^\perp} b = 0$, which holds iff $b \in W$. The equivalence with $\rho_W(b) = 0$ follows from $\|x\| = 0$ iff $x = 0$. □

Proposition 2.5 (Hodge decomposition surface)**.** *Let*

$$d_{\text{next}} d_{\text{prev}} = 0, \qquad \Delta = d_{\text{prev}} d_{\text{prev}}^* + d_{\text{next}}^* d_{\text{next}}.$$

Then every cochain b admits a decomposition

$$b = h + e_1 + e_2$$

with

$$\Delta h = 0, \qquad e_1 \in \operatorname{im}(d_{\text{prev}}), \qquad e_2 \in \operatorname{im}(d_{\text{next}}^*),$$

and

$$\langle h, e_1 \rangle = \langle h, e_2 \rangle = \langle e_1, e_2 \rangle = 0.$$

Proof. Since Δ is self-adjoint, decompose the ambient cochain space as

$$\ker(\Delta) \oplus \operatorname{im}(\Delta).$$

Since

$$\operatorname{im}(\Delta) \subseteq \operatorname{im}(d_{\text{prev}}) + \operatorname{im}(d_{\text{next}}^*),$$

every cochain b can be written as $h + e_1 + e_2$ with $h \in \ker \Delta$, $e_1 \in \operatorname{im}(d_{\text{prev}})$, and $e_2 \in \operatorname{im}(d_{\text{next}}^*)$. From $\Delta h = 0$, one obtains

$$\|d_{\text{prev}}^* h\|^2 + \|d_{\text{next}} h\|^2 = 0,$$

hence $d_{\text{prev}}^* h = 0$ and $d_{\text{next}} h = 0$. These identities give $\langle h, e_1 \rangle = 0$ and $\langle h, e_2 \rangle = 0$, while $d_{\text{next}} d_{\text{prev}} = 0$ gives $\langle e_1, e_2 \rangle = 0$. □

2.1 Public And Hidden Data

Definition 2.6 (Public observation)**.** *A public observation is the data exposed to a verifier or adversary before choosing a hidden representative. In the abstract model it consists of a public class or shadow value p, together with a cochain-facing observation b. In the Homotopical Encrypt instantiation, the fully public handles are hashes, roots, topology hashes, nullifiers, and bounded public summaries; cochain commitments and proof-carrying cochain payloads sit in the verifier-facing artifact layer; full cochains and openings remain witness data unless an explicit protocol path exposes them.*

The public observation is not the hidden Hodge system. It classifies a public equivalence class of possible hidden representatives. Fibrational and homotopical semantics enter through this distinction: public equality is equality after applying a public classifier, not strict equality of hidden representatives. Section 4 develops this semantic layer.

Definition 2.7 (Hidden-operator public projection)**.** *A public projection is a map*

$$\pi : \{\textit{hidden Hodge operators } \Delta\} \longrightarrow P$$

to a public class type P. Two hidden operators are publicly equivalent when $\pi(\Delta_1) = \pi(\Delta_2)$. The hidden operator fiber over $p \in P$ is

$$\mathcal{H}_p = \pi^{-1}(p) = \{\Delta : \pi(\Delta) = p\},$$

and the hidden class of Δ is $\mathcal{H}_{\pi(\Delta)}$.

Definition 2.8 (Lossiness and kernel ambiguity)**.** *The public projection π is* lossy *if there exist $\Delta_1 \neq \Delta_2$ with $\pi(\Delta_1) = \pi(\Delta_2)$. The two operators have the same harmonic kernel if*

$$\forall v, \quad \Delta_1 v = 0 \Longleftrightarrow \Delta_2 v = 0.$$

The projection is kernel-ambiguous *if there are publicly equivalent operators Δ_1, Δ_2 whose harmonic kernels are not the same.*

Definition 2.9 (Public-compatible Hodge ambiguity)**.** *Public-compatible Hodge ambiguity means that there exist publicly equivalent operators Δ_1, Δ_2 and a cochain-facing observation b such that*

$$\Delta_1 b = 0 \qquad \textit{but} \qquad \Delta_2 b \neq 0.$$

Proposition 2.10 (Ambiguity hierarchy)**.** *Public-compatible Hodge ambiguity implies kernel ambiguity, and kernel ambiguity implies that the public projection is lossy.*

Proof. If b is harmonic for Δ_1 but not for Δ_2, then the two operators cannot have the same harmonic kernel. This gives kernel ambiguity. If publicly equivalent operators have different kernels, they cannot be the same operator, since equality of operators implies equality of kernels. Therefore the public projection identifies two distinct hidden operators. □

Definition 2.11 (Hidden Hodge witness)**.** *A hidden Hodge witness for a public observation (p, b) consists of hidden operator data Δ, a harmonic cochain h, and Hodge-structured noise e, such that*

$$\Delta \in \mathcal{H}_p, \qquad h \in \ker(\Delta), \qquad e \in \operatorname{im}(\delta) \oplus \operatorname{im}(\delta^*), \qquad b = h + e.$$

The hidden data includes the fibration representative, restriction maps, the induced Laplacian, its harmonic kernel, and private witness/cochain data. Public data may bind to this hidden data through commitments or proof-carrying artifacts, but it does not present the operative kernel.

Definition 2.12 (HCP solution)**.** *A solution to an HCP instance* (p, b) *is a cochain* h' *that is the harmonic component selected by the operative hidden witness. In the exact recovery form used by the hidden-operator separation,* $h' = h$ *for every hidden witness* (Δ, h, e) *compatible with the public observation; hence uniform public recovery requires the observation to determine the relevant hidden harmonicity data.*

2.2 Recovery Target

The recovery target is the harmonic cochain selected by the operative kernel, not the entire kernel subspace. Thus CVP asks for a closest vector in a presented lattice, while HCP asks for a harmonic cochain in the operative Hodge kernel.

In the metric variant used for comparison with CVP, the target is the closest harmonic cochain to b. Once Δ or a basis for $\ker(\Delta)$ is explicitly presented, this becomes a closest-vector question on the harmonic lattice. The hidden-operator model withholds precisely that presentation.

2.3 Computational Model

The computational model separates three layers.

(i) The mathematical layer defines public projections, hidden operator fibers, harmonicity, and kernel ambiguity.

(ii) External hardness assumptions and cited coding-theory theorems enter at explicitly labeled theorem boundaries.

(iii) Runtime obligations such as byte parsing, canonical encodings, hash domain separation, proof verification, and state transition are outside the mathematical HCP definition.

This paper treats the first two layers. The runtime layer belongs to the subsequent protocol paper.

Lean Correspondence

The definitions and theorem surfaces in this section correspond to `Hardness.MatrixIsHarmonic`, `Hardness.PublicProjection`, `Hardness.HiddenOperatorFiber`, `Hardness.HiddenOperatorClass`, `Hardness.SameKernel`, `Hardness.KernelAmbiguous`, `Hardness.PublicCompatibleHodgeAmbiguity`, `Hardness.Hodge_Decomposition_`

`Theorem`, `Hardness . kernelAmbiguous _ implies _ lossy`, `Hardness . kernelAmbiguous _ of _ publicCompatibleHodgeAmbiguity`, and `Hardness.Hodge_Ambiguity`.

3 CVP Comparison And Kernel Ambiguity

CVP asks for a closest vector in an explicitly presented lattice. HCP asks for the harmonic cochain selected by a hidden Hodge operator. In the Homotopical Encrypt model, public data need not determine the corresponding harmonic kernel.

3.1 Known-Lattice CVP Surface

An explicit-lattice CVP instance consists of a generator matrix B and a target vector t. The presented lattice is

$$\Lambda_B = \{Bx : x \in R^n\}.$$

A correct CVP answer is a vector $v \in \Lambda_B$ such that

$$\operatorname{dist}(t, v) \leq \operatorname{dist}(t, w) \qquad \text{for every } w \in \Lambda_B.$$

Equivalently, for a CVP oracle O,

$$\mathsf{CVP}_{\mathrm{corr}}(O) \iff \forall B, t,\ \Big(O(B,t) \in \Lambda_B \wedge \forall w \in \Lambda_B,\ \operatorname{dist}(t, O(B,t)) \leq \operatorname{dist}(t, w)\Big).$$

The weaker output condition is merely lattice membership:

$$\forall B, t,\ O(B,t) \in \Lambda_B.$$

Fix a class $\mathcal{C}$ of admissible algorithms, such as polynomial-time or probabilistic polynomial-time algorithms. The external CVP hardness assumption is the schema

$$\mathsf{CVP}_{\mathrm{hard}}(\mathcal{C}) \iff \neg\exists O,\ O \in \mathcal{C} \wedge \mathsf{CVP}_{\mathrm{corr}}(O).$$

This is the separation between the known-lattice interface and the hidden-operator interface: a CVP oracle receives B and t, while a public HCP procedure sees only public class values, hashes, commitments, and verifier-facing artifacts unless extra hidden-operator information is supplied.

3.2 Reduction Direction

Let B present $\Lambda_B = \mathrm{im}(B)$. If a matrix Δ satisfies $\ker(\Delta) = \Lambda_B$, then any solver satisfying

$$\Delta S(t) = 0$$

returns $S(t) \in \Lambda_B$. If, in addition,

$$\forall h, \quad \Delta h = 0 \Rightarrow \mathrm{dist}(t, S(t)) \leq \mathrm{dist}(t, h),$$

then $S(t)$ is closest to t among all vectors of Λ_B.

Proposition 3.1 (Kernel realization for explicit lattices)**.** *Let B be an explicit generator matrix over a field and set $\Lambda_B = \mathrm{im}(B)$. There exists a matrix Δ such that*

$$\ker(\Delta) = \Lambda_B.$$

Consequently, any solver whose output is Δ-harmonic for every presented Δ returns a vector in the explicit lattice Λ_B when run on this realized operator. If that solver is also closest among Δ-harmonic vectors, then its output is a CVP solution for (B, t).

Proof. Let B be an explicit generator matrix for a lattice

$$\Lambda_B = \{Bx : x \in R^n\} \subseteq R^n$$

and let $t \in R^n$ be the CVP target. We construct a matrix Δ with

$$\ker(\Delta) = \mathrm{im}(B) = \Lambda_B.$$

Put $V = \mathrm{im}(B)$. Since R^n is finite-dimensional over a field, choose a complementary subspace W with

$$V + W = R^n, \qquad V \cap W = 0.$$

Thus every $y \in R^n$ has a unique decomposition

$$y = v + w, \qquad v \in V, \quad w \in W.$$

Equivalently, there is a unique element $p(y) \in W$ such that $y - p(y) \in V$. Define

$$P(y) = p(y).$$

The uniqueness of the $V \oplus W$ decomposition gives linearity: for $y, z \in R^n$, the vector $P(y) + P(z)$ lies in W, and

$$(y + z) - (P(y) + P(z)) = (y - P(y)) + (z - P(z)) \in V,$$

so $P(y+z) = P(y) + P(z)$. Similarly, for $a \in R$, $aP(y) \in W$ and

$$ay - aP(y) = a(y - P(y)) \in V,$$

so $P(ay) = aP(y)$.

We now compute the kernel. If $y \in V$, then $0 \in W$ and $y - 0 \in V$, hence uniqueness gives $P(y) = 0$. Conversely, if $P(y) = 0$, then $y - P(y) = y \in V$. Therefore

$$\ker P = V = \mathrm{im}(B).$$

Let Δ be the matrix of the linear map P. Then, for every $y \in R^n$,

$$\Delta y = 0 \quad \Longleftrightarrow \quad y \in \ker P \quad \Longleftrightarrow \quad y \in \mathrm{im}(B) \quad \Longleftrightarrow \quad \exists x,\ Bx = y.$$

Let $S : R^n \to R^n$ be a presented-operator Hodge solver satisfying

$$\forall \Delta\, b, \qquad \Delta(S(b)) = 0.$$

Applying this to the Δ constructed above and $b = t$ gives

$$\Delta(S(t)) = 0.$$

By the kernel calculation, there exists x with

$$Bx = S(t).$$

Hence $S(t) \in \Lambda_B$.

Suppose in addition that S satisfies

$$\forall h', \quad \Delta h' = 0 \Longrightarrow \mathrm{dist}(t, S(t)) \le \mathrm{dist}(t, h').$$

Since the Δ-harmonic vectors are exactly the vectors of Λ_B, this inequality is precisely

$$\mathrm{dist}(t, S(t)) \le \mathrm{dist}(t, w) \qquad \text{for every } w \in \Lambda_B.$$

Together with $S(t) \in \Lambda_B$, this is exactly CVP correctness for the explicit instance (B, t). □

For fixed Δ, closest harmonic recovery is minimization over $\ker(\Delta)$.

Proposition 3.2 (Projection-model Hodge recovery is CVP-shaped)**.** *Let $W \subseteq R^n$, let $P_W : R^n \to W$ be orthogonal projection, and set*

$$\Delta_W = I - P_W.$$

Then

$$\Delta_W h = 0 \quad \Longleftrightarrow \quad h \in W.$$

Hence

$$\Big(\Delta_W h = 0 \ \wedge \ \forall v,\ \Delta_W v = 0 \Rightarrow \mathrm{dist}(b,h) \le \mathrm{dist}(b,v)\Big) \quad \Longleftrightarrow$$
$$\Big(h \in W \ \wedge \ \forall v \in W,\ \mathrm{dist}(b,h) \le \mathrm{dist}(b,v)\Big).$$

Proof.

$$\Delta_W b = (I - P_W)b = P_{W^\perp} b.$$

Hence

$$\Delta_W u = 0 \quad \Longleftrightarrow \quad P_{W^\perp} u = 0 \quad \Longleftrightarrow \quad u \in W$$

for every $u \in R^n$. Substituting this equivalence for $u = h$ gives the harmonicity term on the right. Substituting it for every $u = v$ in the universal quantifier identifies minimization over $\{v : \Delta_W v = 0\}$ with minimization over W. This proves the displayed equivalence. □

3.3 Oracle Insufficiency

The strictness direction uses a different contract. A public Hodge solver may call an explicit-lattice CVP oracle, but it receives no hidden operator. The predicates below encode uniform harmonicity with respect to every hidden operator and fixedness on inputs already harmonic for that operator.

Definition 3.3 (Public-data Hodge solver contract)**.** *Let*

$$S : \mathcal{O}_{\mathrm{CVP}} \times R^n \longrightarrow R^n,$$

where $\mathcal{O}_{\mathrm{CVP}}$ is the class of explicit-lattice CVP oracles. For $O \in \mathcal{O}_{\mathrm{CVP}}$, define

$$\mathsf{H}_O(S) \Longleftrightarrow \forall \text{ linear } \Delta : R^n \to R^n,\ \forall b \in R^n, \quad \Delta(S(O,b)) = 0,$$
$$\mathsf{F}_O(S) \Longleftrightarrow \forall \text{ linear } \Delta : R^n \to R^n,\ \forall b \in R^n, \quad \Delta b = 0 \Rightarrow S(O,b) = b.$$

The public-data-only contract is

$$\exists S, \quad \forall O,\ \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Proposition 3.4 (Public ambiguity and CVP oracle insufficiency)**.** *Suppose* $\dim R^n \ge 2$. *Then:*

(i) The constant public projection on hidden linear Hodge operators is public-compatible Hodge ambiguous, hence kernel-ambiguous and lossy.

(ii) If at least one correct explicit-lattice CVP oracle exists, namely

$$\exists O,\ \mathsf{CVP}_{\mathrm{corr}}(O),$$

then

$$\neg\exists S,\quad \forall O,\ \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow (\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)).$$

Hence a correct oracle for explicit known-lattice CVP does not by itself determine the operative Hodge–Laplacian kernel.

Proof. Assume $\dim R^n \geq 2$, and choose an index i. Let

$$b = e_i, \qquad \Delta_1 = 0, \qquad \Delta_2 = \mathrm{diag}(1, 0, \ldots, 0),$$

where the nonzero diagonal entry of Δ_2 is placed at i. Then

$$\Delta_1 b = 0, \qquad \Delta_2 b = b \neq 0.$$

Thus b is harmonic for Δ_1 but not harmonic for Δ_2. The constant public projection sends every hidden operator to the same public value, so Δ_1 and Δ_2 are publicly equivalent. The same cochain-facing observation b is therefore compatible with two hidden Hodge operators that disagree about whether b lies in the harmonic kernel. This gives public-compatible Hodge ambiguity; the implications of Theorem 2.10 give kernel ambiguity and lossiness.

Now suppose, in addition, that $\exists O, \mathsf{CVP}_{\mathrm{corr}}(O)$. Suppose, for contradiction, that S satisfies the displayed public-data-only contract. Choose O with $\mathsf{CVP}_{\mathrm{corr}}(O)$. Then

$$\mathsf{H}_O(S) \wedge \mathsf{F}_O(S).$$

Since $\Delta_1 b = 0$, the predicate $\mathsf{F}_O(S)$, applied to (Δ_1, b), gives

$$S(O, b) = b.$$

The predicate $\mathsf{H}_O(S)$, applied to (Δ_2, b), gives

$$\Delta_2 S(O, b) = 0.$$

Combining the two equalities gives $\Delta_2 b = 0$, contradicting the construction of b. Hence no such public-data-only solver exists, even assuming the availability of a correct oracle for explicitly presented CVP instances. □

The obstruction is kernel ambiguity: the same public observation may be compatible with multiple hidden operators whose harmonic kernels differ. A CVP oracle can solve a closest-vector problem after the lattice is presented; it does not select which hidden kernel public data was meant to present.

3.4 Structural Comparison

The comparison separates three problems:

(i) known-lattice closest-vector search;

(ii) hidden-operator HCP;

(iii) public projection/proximity verification.

The first two carry the hardness comparison. The third is verifier-facing: given accepted hidden/geometric data, later sections construct finite public shadows and an RS/MCA endpoint without claiming that the projection is injective.

Lean Correspondence

The explicit-lattice comparison corresponds to `CVP.ExplicitLattice`, `CVP.CVP_Instance`, `CVP.CVPSolution`, `CVP.CVPOracleCorrect`, `CVP.CVPHardnessAssumption`, `Hardness.exists_matrix_ker_eq_range`, `Hardness.Hodge_Hardness_Reduction_Proof`, `Hardness.HodgeCVPSolution`, `Hardness.ProjectionHodgeCVPSolution`, `Hardness.matrix_harmonic_iff_projection_isHarmonic`, `Hardness.Hodge_Projection_Is_CVP`, `Hardness.no_public_solver_on_ambiguous_input`, and `Hardness.CVP_Oracle_Insufficient_For_Public_Hodge_Solving`.

4 Axioms, Semantics, And Public Data

4.1 Fibrational Semantics

The semantic layer is motivated by Helfer's first-order homotopical logic [Hel19]. We use the associated path, fibration, sheaf, and internal-logic language for hidden representatives, public bases, and transport of witness data; the external foundational results are taken as background rather than reproved. We use the usual fibrational notation

$$p : E \longrightarrow B, \qquad E_b = p^{-1}(b)$$

for a total space of hidden data over a base of public or contextual data. When $x, y \in E_b$, equality in the semantic layer is interpreted through a path

object rather than by strict equality in the ambient presentation. We write $\mathcal{P}_X(x, y)$ for the path object from x to y in a space or type X, and set

$$x \simeq_b y \quad \Longleftrightarrow \quad \mathcal{P}_{E_b}(x, y) \neq \varnothing.$$

For a dependent family $F : E \to \mathsf{Type}$, a path $\gamma \in \mathcal{P}_E(x, y)$ supplies the transport map

$$\gamma_* : F_x \longrightarrow F_y.$$

Thus the primitive semantic data are fibers, paths, and transport along paths. The cryptographic objects introduced later live over this interface; they do not replace it with byte-level equality.

We also use the corresponding categorical fibration vocabulary. For a functor $p : E \to B$, a morphism $f : U \to V$ in E is p-Cartesian when every arrow $g : W \to V$ and base arrow $\alpha : p(W) \to p(U)$ satisfying

$$p(g) = \alpha\, p(f)$$

factor uniquely through f. A Grothendieck fibration is then a functor with Cartesian lifts for every base arrow. This is the categorical form of the transport principle used by the hidden-representative semantics.

Proposition 4.1 (Fibration copies preserve hidden semantics)**.** *Let $p : E \to B$ be a Grothendieck fibration. For each tag n, let $E^{(n)}$ be a tagged copy of E and let $p^{(n)} : E^{(n)} \to B$ be the induced fibration. Then $p^{(n)}$ is again a Grothendieck fibration, and p and $p^{(n)}$ are homotopy equivalent over B.*

Proof. The tagged copy has the same objects and morphisms as E, with only the external tag changed. The forgetful functor $E^{(n)} \to E$ and the tagging functor $E \to E^{(n)}$ compose to the identity up to the evident natural isomorphisms. Cartesian lifts in $E^{(n)}$ are transported directly from Cartesian lifts in E, so the induced functor remains a Grothendieck fibration. The two functors give the homotopy equivalence over the base B. □

4.2 Fibrational Identity Objects

Let $\Delta_A : A \to A \times A$ denote the diagonal. Following the path/diagonal semantics, the identity object over a pair (x, y) has fiber $\mathcal{P}_A(x, y)$. We therefore regard x and y as equal for the first-order homotopical semantics exactly when this fiber is inhabited:

$$x \simeq_A y \quad \Longleftrightarrow \quad \mathcal{P}_A(x, y) \neq \varnothing.$$

This is the semantic source of the hidden-operator model used in HCP: the verifier may receive evidence that two representatives lie in the same homotopical identity class without receiving the representative or the path witness itself.

4.3 Homotopy Classes

Let A be a type of hidden representatives and let P be a type of public class data. A public classifier is a map

$$\pi : A \longrightarrow P.$$

It determines public fibers

$$A_p = \pi^{-1}(p) = \{x \in A : \pi(x) = p\}$$

and a public equivalence relation

$$x \equiv_\pi y \quad \Longleftrightarrow \quad \pi(x) = \pi(y).$$

The classifier is path-respecting when

$$x \simeq_A y \Longrightarrow x \equiv_\pi y,$$

and non-faithful when some public fiber contains distinct hidden representatives. A hidden class-membership witness is a path $\gamma \in \mathcal{P}_A(x, y)$; its public shadow is only the equality $\pi(x) = \pi(y)$. At this stage the shadow is only a classifier or fingerprint. The finite-code projection and Reed–Solomon/MCA endpoint are introduced later in Section 6.

4.4 Axiom And Interface Surfaces

Here a *surface* means a mathematical assumption, specification predicate, or interface boundary used by the paper. It does not mean that every item below is literally a file under `Axioms/`. The relevant surfaces are the following.

(i) Foundational cochain/Hodge surface: a cochain complex, a coboundary, the induced Hodge–Laplacian, Hodge decomposition, and orthogonality.

(ii) Hidden semantic surface: paths, transport, fibration structure, homotopy classes, public classifiers, and the nonfaithfulness of public class data.

(iii) Sheaf-coherence surface: restriction-shape compatibility, naturality, $d^2 = 0$, the induced Hodge–Laplacian identity, finiteness, simpliciality, and connectedness.

(iv) Spectral and stage-admissibility surface: Tsirelson-style spectral window predicates, near-boundary predicates, and object-local admissibility wrappers for candidate Hodge–Laplacian data.

(v) Public-judgment surface: finite Heyting collapse, public truth equivalence, oracle query/answer predicates, and oracle-visible spectral, identity, and topology judgments.

(vi) Public-artifact surface: cochain-commitment payloads, commitment security assumptions, proof-system assumptions, proof-carrying artifacts, and compression/observation boundaries used in Section 5.

Only the well-formedness boundaries that are directly consumed later are displayed as conjunctions below. The path/fibration, Heyting/oracle, spectral, stage, and runtime objects are invoked through their stated interface predicates unless a particular object-level predicate is displayed.

Definition 4.2 (Foundation assumption boundary)**.** *A foundation record is well formed exactly when*

$$\begin{gathered}\mathsf{CochainComplex} \wedge \mathsf{CoboundaryLinearity} \\ \wedge \mathsf{LaplacianFromCoboundary} \wedge \mathsf{HodgeDecomposition} \\ \wedge \mathsf{HodgeOrthogonality} \\ \wedge \mathsf{CommitmentBinding} \wedge \mathsf{CommitmentHiding} \\ \wedge \mathsf{CommitmentHomomorphic}.\end{gathered}$$

Definition 4.3 (Sheaf-coherence assumption boundary)**.** *A sheaf-coherence record is well formed exactly when*

$$\begin{gathered}\mathsf{IncidenceShapes} \wedge \mathsf{EdgeSquareCommutes} \\ \wedge \mathsf{TriangleFacesCommute} \wedge \mathsf{CoboundarySquaresZero} \\ \wedge \mathsf{LaplacianFromCoboundary} \wedge \mathsf{Orthogonality} \\ \wedge \mathsf{Finite} \wedge \mathsf{Simplicial} \wedge \mathsf{Connected}.\end{gathered}$$

Definition 4.4 (Proof-system and commitment-security boundaries)**.** *The commitment-security boundary is*

$$\begin{gathered}\mathsf{Binding} \wedge \mathsf{Hiding} \wedge \mathsf{HomomorphicAddition} \wedge \mathsf{HomomorphicScalarMult} \\ \wedge \mathsf{TopologyBinding} \wedge \mathsf{PartialOpeningSoundness}.\end{gathered}$$

The proof-system boundary is

$$\mathsf{Completeness} \wedge \mathsf{Soundness} \wedge \mathsf{ZeroKnowledge} \wedge \mathsf{StatementBinding}.$$

Consensus, ledger, reliability, and Proof-of-Value records are mentioned here only at their mathematical interface boundary. Their runtime policy content belongs to the subsequent protocol paper.

4.5 Exact Structural Invariants

The exact structural invariants are not a separate source of geometry. They package the same foundational and sheaf-coherence hypotheses at the point where an object is admitted. We use the cellular-sheaf Hodge–Laplacian viewpoint of Hansen–Ghrist as background [HG18]. For a cochain complex $C^\bullet$, the accepted data supply a differential

$$d^k : C^k \longrightarrow C^{k+1}$$

with $d^{k+1}d^k = 0$, together with an adjoint d^* and the induced Hodge Laplacian

$$\Delta = d^*d + dd^*.$$

In the sheaf setting, the corresponding coherence assumptions assert that restriction maps, local data, and gluing data are compatible with this cochain/Hodge structure. The only structural consequences used later are the differential identity $d^2 = 0$ and the Hodge–Laplacian identity above. Equivalently, the projection maps used to pass from hidden sheaf data to public cochain-facing data are required to commute with the coboundary:

$$\begin{array}{ccc} F(V) & \xrightarrow{\eta_V} & G(V) \\ {\scriptstyle \delta_F}\downarrow & & \downarrow{\scriptstyle \delta_G} \\ F(E) & \xrightarrow{\eta_E} & G(E) \end{array} \qquad \eta_E \circ \delta_F = \delta_G \circ \eta_V.$$

This is the functorial/naturality diagram used in the paper. It is a cochain-level coherence condition, not a runtime statement about byte transport or consensus execution.

Proposition 4.5 (Exact structural preconditions)**.** *Accepted sheaf-coherence data supply*

$$d^2 = 0 \qquad \textit{and} \qquad \Delta = d^*d + dd^*$$

for the cochain complex under discussion.

4.6 Spectral-Gap Admissibility

The spectral condition is an object-level admissibility predicate. It is not the source of HCP hardness, not a reconstruction theorem for the hidden fibration, and not introduced here as a consensus rule. In Paper 1 the PoV/Tsirelson spectral window is a mathematical predicate on candidate Hodge–Laplacian data; its use as a protocol-level admissibility rule belongs to the subsequent protocol paper. The terminology is Tsirelson-style because the admissible window uses the familiar $\sqrt{2}$ scale from Tsirelson's bound [Tsi80]. For a Hermitian positive semidefinite operator Δ, write

$$\lambda_1^+(\Delta) = \inf\{\lambda : \lambda \text{ is an eigenvalue of } \Delta,\ \lambda \neq 0\},$$

with $\lambda_1^+(\Delta) = 0$ when Δ has no nonzero spectral value. The Tsirelson-style admissibility axiom is the spectral window

$$\frac{\sqrt{2}}{2} \leq \lambda_1^+(\Delta) \leq 2\sqrt{2}.$$

A separate boundary predicate marks Δ as ε-near one of the two endpoints:

$$\left|\lambda_1^+(\Delta) - \frac{\sqrt{2}}{2}\right| \leq \varepsilon \quad \text{or} \quad \left|\lambda_1^+(\Delta) - 2\sqrt{2}\right| \leq \varepsilon.$$

This predicate records near-boundary status only; runtime policy for such values belongs to the subsequent protocol paper.

The corresponding Grassmannian/Laplacian surface makes the hidden-kernel search space concrete. Let

$$\mathcal{L}_\beta(R^n) = \{\Delta : \Delta = \Delta^*,\ \Delta \succeq 0,\ \dim_R \ker \Delta = \beta\}$$

be the Laplacian space at kernel dimension β. For $W \in \mathrm{Gr}(k, R^n)$, let P_W be the orthogonal projection onto W and set

$$\Delta_W = I - P_W.$$

Thus $\ker(\Delta_W) = W$, and the Grassmannian family is

$$\Omega_k = \{\Delta_W : W \in \mathrm{Gr}(k, R^n)\}.$$

Equivalently, a point of $\mathrm{Gr}(k, R^n)$ is a k-dimensional harmonic subspace W, and the map $W \mapsto \Delta_W$ sends it to the corresponding projection Laplacian. In particular, Ω_1 is the one-dimensional Grassmannian family of rank-one harmonic kernels.

Proposition 4.6 (Grassmannian Laplacian search space)**.** *For every $k \leq n$, the family Ω_k is nonempty and satisfies*

$$\Omega_k \subseteq \{\Delta_W : W \subset R^n \text{ a linear subspace}\}.$$

Moreover, every element of Ω_k has a k-dimensional harmonic kernel, so $\Omega_k \subseteq \mathcal{L}_k(R^n)$.

Proof. If $k \leq n$, choose k coordinate basis vectors in R^n. Their span is a k-dimensional subspace W, and $\Delta_W \in \Omega_k$, proving nonemptiness. The displayed containment is immediate from the definition of Ω_k. Since $\Delta_W = I - P_W = P_{W^\perp}$, its kernel is exactly W, hence has dimension k. □

Proposition 4.7 (Injectivity of the Grassmannian Laplacian map)**.** *If $W_1, W_2 \subset R^n$ are subspaces and*

$$\Delta_{W_1} = \Delta_{W_2},$$

then $W_1 = W_2$. Consequently $W \mapsto \Delta_W$ is injective on $\mathrm{Gr}(k, R^n)$.

Proof. The equality $\Delta_{W_1} = \Delta_{W_2}$ is the equality $I - P_{W_1} = I - P_{W_2}$, hence $P_{W_1} = P_{W_2}$. For any $x \in W_1$, $P_{W_1}x = x$, so $P_{W_2}x = x$, and therefore $x \in W_2$. Thus $W_1 \subseteq W_2$. The reverse containment is identical, so $W_1 = W_2$. □

The higher-rank Grassmannian search spaces contain explicit one-parameter families. Fix $1 \leq k < n$. Choose linearly independent vectors $u, v \in R^n$ and a $(k-1)$-dimensional subspace W_0 disjoint from $\mathrm{span}\{u, v\}$. For $t \in R$, set

$$W(t) = \mathrm{span}\{u + tv\} + W_0.$$

Then $W(t)$ has dimension k, and varying t gives a corresponding family

$$\Delta_{W_t} = I - P_{W_t}.$$

Proposition 4.8 (Parametric Laplacian search space)**.** *If $1 \leq k < n$, then there exist u, v, W_0 as above such that the map*

$$R \longrightarrow \mathrm{Gr}(k, R^n), \qquad t \longmapsto W(t)$$

is injective. Hence $t \mapsto \Delta_{W(t)}$ is an injective one-parameter family in Ω_k.

Proof. The existence of u, v, W_0 follows by choosing a $(k-1)$-plane and then two independent directions outside it; the inequality $k < n$ leaves enough room for this choice. The dimension computation uses the disjointness:

$$\dim W(t) = \dim \mathrm{span}\{u + tv\} + \dim W_0 = 1 + (k-1) = k.$$

If $W(t_1) = W(t_2)$, then $u + t_1 v$ lies in $\mathrm{span}\{u + t_2 v\} + W_0$. Modulo W_0, this gives a linear dependence between u and v unless $t_1 = t_2$. Since u, v were chosen independent and W_0 was disjoint from their span, $t_1 = t_2$. Injectivity of $t \mapsto \Delta_{W(t)}$ follows from Theorem 4.7. □

Proposition 4.9 (Projection spectrum and spectral gap)**.** *For every $W \subset R^n$, the projection Laplacian $\Delta_W = I - P_W = P_{W^\perp}$ has spectrum contained in $\{0, 1\}$. Its kernel is W, its image is $W^\perp$, and, if $W \neq R^n$, its least nonzero eigenvalue and largest eigenvalue are both 1. Thus the relative spectral gap of the Grassmannian projection family is 1.*

Proof. If $x \in W$, then $\Delta_W x = 0$. If $x \in W^\perp$, then $\Delta_W x = x$. Since $R^n = W \oplus W^\perp$, these two subspaces exhaust the eigenspace decomposition. Therefore the only possible eigenvalues are 0 and 1, with 0-eigenspace W and 1-eigenspace $W^\perp$. When $W^\perp \neq 0$, the nonzero spectral values consist only of 1, so both λ_1^+ and $\lambda_{\max}$ are 1. □

The likelihood surface records the corresponding adversary vocabulary. A classical adversary is a map from an observation b to a guessed Laplacian; a quantum adversary is a probability density on candidate Laplacians. For a noise scale σ, the Gaussian likelihood of a candidate A at observation b is written abstractly as

$$L_\sigma(b, A) = \exp\left(-\frac{\|Ab\|_2^2}{2\sigma^2}\right),$$

with normalized classical and quantum success probabilities obtained by integrating or maximizing this likelihood over the chosen family Ω_k. These likelihood objects are comparison surfaces for quantitative security statements; they are not assumptions used to prove the hidden-operator separation in Theorem 3.4.

Definition 4.10 (Adversary likelihood surfaces)**.** *A classical adversary on R^n is a function*

$$\mathcal{A}_{\mathrm{cl}} : R^n \to M_n(R)$$

from an observation to a guessed Laplacian. A quantum adversary on a candidate family Ω_k is a nonnegative density

$$\mathcal{A}_{\mathrm{q}} : R^n \times \Omega_k \to R_{\geq 0}$$

whose integral over Ω_k is 1 for each observation. With the standard Gaussian prior on observations, the corresponding normalized success surfaces are

obtained from

$$\int_{R^n} L_\sigma(b, \mathcal{A}_{\mathrm{cl}}(b)) e^{-\|b\|_2^2/2}\, db$$

in the classical case and

$$\int_{R^n} \left(\int_{\Omega_k} \mathcal{A}_{\mathrm{q}}(b, A) L_\sigma(b, A)\, dA \right) e^{-\|b\|_2^2/2}\, db$$

in the quantum-density case, up to the usual Gaussian normalization factor.

The likelihood is written with the Euclidean norm. The norm-convention check is that the default function-space norm may be the supremum norm, so the Euclidean likelihood must be represented by passing to the ℓ^2 norm explicitly. The mathematical surface used here is the corrected Euclidean one.

Proposition 4.11 (Likelihood for a projection Laplacian)**.** *For $A = \Delta_W$, the Euclidean Gaussian likelihood is*

$$L_\sigma(b, \Delta_W) = \exp\left(-\frac{\|P_{W^\perp} b\|_2^2}{2\sigma^2} \right).$$

Proof. By Theorem 2.4, $\Delta_W b = P_{W^\perp} b$. Substitute this identity into the definition of $L_\sigma(b, A)$. □

Proposition 4.12 (Single-observation optimal classical guess)**.** *If $b \neq 0$ and $W_b = \mathrm{span}\{b\}$, then*

$$L_\sigma(b, \Delta_{W_b}) = 1.$$

Equivalently, the projection residual of b against the Laplacian whose kernel is $\mathrm{span}\{b\}$ is zero.

Proof. Since $b \in W_b$, $\Delta_{W_b} b = 0$. Hence $L_\sigma(b, \Delta_{W_b}) = \exp(0) = 1$. □

This proposition is deliberately local: it concerns one observation b and the candidate subspace generated by that same observation. It does not collapse the HCP search problem, because the hardness comparison ranges over the whole hidden Grassmannian family and over public observations that do not reveal the operative kernel.

Proposition 4.13 (Gaussian prior factorization)**.** *For $x = (x_i)_{i=1}^n \in R^n$,*

$$\exp\left(-\frac{\|x\|_2^2}{2} \right) = \prod_{i=1}^n \exp\left(-\frac{x_i^2}{2} \right).$$

Proof. Since $\|x\|_2^2 = \sum_i x_i^2$, the left hand side is

$$\exp\left(\sum_i -\frac{x_i^2}{2}\right),$$

which equals the product of the exponentials. □

Definition 4.14 (Spectral fingerprint)**.** *A finite-dimensional sheaf Laplacian is a self-adjoint positive semidefinite linear operator $L : V \to V$. Its spectral fingerprint is the multiset*

$$\mathrm{Spec}_{\mathrm{multi}}(L)$$

of eigenvalues counted with multiplicity. Two Laplacians are spectrally equivalent when these multisets agree. In contrast, the coarse homotopy equivalence surface used here records only harmonic dimension:

$$L_1 \simeq_{\ker} L_2 \quad \Longleftrightarrow \quad \dim\ker L_1 = \dim\ker L_2.$$

Proposition 4.15 (Spectrum is not determined by harmonic dimension)**.** *There exist self-adjoint positive semidefinite Laplacians L_1, L_2 with*

$$\dim\ker L_1 = \dim\ker L_2$$

but

$$\mathrm{Spec}_{\mathrm{multi}}(L_1) \neq \mathrm{Spec}_{\mathrm{multi}}(L_2).$$

Proof. Take $V = R$, $L_1 = \mathrm{id}_R$, and $L_2 = 2\,\mathrm{id}_R$. Both kernels are zero, so the harmonic dimensions agree. The spectral fingerprints are the singleton multisets $\{1\}$ and $\{2\}$, respectively, so the operators are not spectrally equivalent. □

Proposition 4.16 (Spectral-gap admissibility)**.** *If a candidate Hodge–Laplacian geometry carries a by-construction PoV/Tsirelson admissibility witness, then*

$$\lambda_1^+(\Delta) \in \left[\frac{\sqrt{2}}{2},\, 2\sqrt{2}\right].$$

Thus the by-construction witness is an admissibility certificate for the displayed spectral window. It constrains which candidate objects are admissible; it does not remove hidden-operator ambiguity. The HCP hardness mechanism in Section 2 remains the hidden fibration, hidden Laplacian, and hidden kernel.

For a nonempty family $\Omega \subseteq \mathcal{L}_1(R^n)$, the associated classical and quantum adversary-bound forms are spectral-statistical targets, not HCP hardness assumptions. The corresponding empty-family statements are only boundary checks; they are not nontrivial hardness bounds.

4.7 Heyting And Oracle Judgment Surface

The ownership split is:

$$\begin{aligned} \text{hidden equality} &: x \simeq y \Longleftrightarrow \mathcal{P}(x,y) \neq \varnothing, \\ \text{public class data} &: x \equiv_\pi y \Longleftrightarrow \pi(x) = \pi(y), \\ \text{public truth data} &: q : \Omega \longrightarrow \Omega_3. \end{aligned}$$

The first line is the Helfer path/diagonal interface, together with the usual sheaf-theoretic interpretation of intuitionistic logic [Hel19]. The second line is the public classifier surface defined above. The third line is the finite public judgment surface. We take

$$\Omega_3 = \{\bot, ?, \top\}, \qquad \bot < ? < \top,$$

with meet and join given by minimum and maximum in the chain. Its implication is the residuated operation characterized by

$$c \leq (a \Rightarrow b) \quad \Longleftrightarrow \quad a \wedge c \leq b.$$

The quotient map q is not assumed faithful:

$$\exists\, \omega_1 \neq \omega_2 \quad \text{with} \quad q(\omega_1) = q(\omega_2).$$

The oracle-specific component is the query/answer mechanism that produces these public judgments. For this boundary we use oracle-modality terminology compatible with Ahman–Bauer [AB26]: oracle-visible predicates such as vertex compatibility, Betti-number compatibility, public identity-hash compatibility, and spectral compatibility are evaluated in the finite public Heyting surface. Thus an oracle answer is a public quotient of hidden semantic evidence, not a reconstruction of the hidden representative, hidden path witness, fibration, Laplacian, or harmonic kernel.

Operational maps out of Ω_3, such as protocol actions attached to the intermediate public judgment ?, belong to the subsequent protocol paper's runtime and consensus semantics. They are not the Heyting implication operation and do not alter the residuation law above.

The identity hash used by the oracle interface is an observation obtained from the public Hodge projection and quantization path. It is not a cochain commitment payload. The corresponding stability assumption says that equal public identity inputs give the same identity hash; it is a determinism condition for public identity observations, not a uniqueness or binding theorem for cochain commitments. Similarly, topological observations such

as Betti-number compatibility are homotopy-invariant public data, whereas spectral fingerprints are not determined by homotopy class alone by Theorem 4.15. The oracle interface therefore keeps Betti/topology observations, spectral admissibility, identity hashes, and cochain commitments as separate public predicates.

This modular split is part of the mathematical statement: cryptographic assumptions, public classifiers, and Hodge-theoretic claims are not identified. Exact and oracle invariants are used here as mathematical acceptance predicates; their full CS/protocol treatment is deferred to the subsequent protocol paper.

Lean Correspondence

The semantic and axiom surfaces in this section correspond to `Axioms.HelferFOL`, `Axioms.Fibration`, `Axioms.HomotopyClass`, `Axioms.Foundation`, `Axioms.FoundationWellFormed`, `Axioms.SheafCoherence`, `Axioms.SheafCoherenceWellFormed`, `Gates.Exact`, `Gates.structural_preconditions_of_sheaf_exact_gates`, `Axioms.SpectralBounds`, `Tsirelson`, `Gates.Spectral`, `Axioms.StageAdmissibility`, `Gates.Stage`, `Axioms.Heyting`, and `Axioms.OracleModality`.

5 Public Artifact Stack And Lossy Projection

Commitments and public shadows are the first concrete public-facing objects in the paper. They explain what public data can carry without revealing the hidden Hodge representative. They are the scaffolding for transparency in action and anonymity in identity: actions are exposed through auditable public artifacts, while identities remain inside hidden representatives and public equivalence classes.

5.1 Public, Artifact, And Private Layers

The public-facing stack has three mathematical levels. The fully public handle level contains hashes, roots, topology hashes, nullifiers, public summaries, and oracle judgments. The verifier-facing artifact level contains public shadows, cochain commitments, proof-carrying cochain payloads, and certificate data. The private witness level contains the fibration representative, transport/restriction data, full cochains and openings, and the Hodge operator whose kernel selects the harmonic component. The paper's claims are stated so that public handles and verifier-facing artifacts bind to hidden witnesses

without turning the hidden witness into public data. This is compatible with the sheaf-theoretic attacker-observation viewpoint; the associated security games and runtime observation model belong to the subsequent protocol paper. The present paper uses only the mathematical public/private separation described below.

5.2 The Artifact Stack

The public artifact stack has the following shape:

hidden/private object
$\to$ public shadow or compressed public datum
$\to$ commitment/root/hash artifact
$\to$ proof-carrying artifact
$\to$ verifier judgment.

Each layer binds to or summarizes the preceding layer. A hash does not replace the mathematical object, a commitment does not replace the witness, and a proof artifact does not replace the proof obligation. The purpose of the stack is to let public verification refer to hidden sheaf/cochain data without exposing it. In this sense, the stack separates what must be publicly accountable from what must remain representative-private.

Hashes serve as public names, integrity bindings, and fixed-size pointers for larger artifacts. A topology hash names the public combinatorial support; a root commits to a batch or storage view; a transcript hash domain-separates a proof statement. None of these hashes asserts validity by itself. Validity comes from the predicate or proof attached to the hashed object.

5.3 Cochain Commitments

Cochain commitments provide the public handle on hidden cochain data. The canonical payload is a mathematical object with a modulus, edge values, declared edge count, and topology hash.

5.4 Commitment Map

Definition 5.1 (Cochain commitment payload)**.** *A cochain commitment payload is a tuple*

$$(q, n, c, \tau)$$

consisting of a modulus q, a declared edge count n, a list c of integer edge values, and a topology hash τ.

The intended commitment map is the transaction-facing coboundary map

$$(m, r) \longmapsto \delta(m \| r) \in C^1 \pmod{q},$$

stored together with the topology hash of the supporting complex. The payload therefore preserves the accepted modulus, the edge-vector shape, the public topology binding, and the serialized C^1-level commitment values. It does not expose the hidden fibration representative, private transport data, the operative Hodge–Laplacian, or the harmonic kernel.

Definition 5.2 (Commitment well-formedness)**.** *A cochain commitment payload is well formed when its modulus is the supported canonical modulus, its declared edge count agrees with the actual edge-value list length, and the edge list is nonempty.*

Proposition 5.3 (Well-formed committed payloads)**.** *A well-formed cochain commitment payload has nonempty edge data and modulus greater than one.*

Proposition 5.4 (Topology-compatible aggregation)**.** *Commitment aggregation is only meaningful for payloads with compatible modulus, declared edge count, and topology hash. This is the public topology-binding condition consumed by the protocol layer.*

The security interface for this commitment layer consists of binding, hiding, homomorphic addition, homomorphic scalar multiplication, topology binding, and partial-opening soundness. The blinded-commitment, vector-commitment, and cohomological-MAC interfaces are separate axiom records rather than consequences silently bundled into the payload shape. In symbols, their well-formedness conditions are:

$$\begin{aligned}
\mathsf{BlindedCommitmentWF} &\iff \begin{cases} \mathsf{HodgeOrthogonality}, \\ \mathsf{DeterministicBasis}, \\ \mathsf{Binding}, \\ \mathsf{Hiding}; \end{cases} \\
\mathsf{VectorCommitmentWF} &\iff \begin{cases} \mathsf{Completeness}, \\ \mathsf{Soundness}, \\ \mathsf{PositionBinding}; \end{cases} \\
\mathsf{CohomologicalMACWF} &\iff \begin{cases} \mathsf{Correctness}, \\ \mathsf{Unforgeability}, \\ \mathsf{KeyBinding}, \\ \mathsf{Determinism}. \end{cases}
\end{aligned}$$

Here HodgeOrthogonality denotes the orthogonal Hodge splitting

$$C^1 = \mathcal{H}^1 \oplus \mathrm{im}(\delta) \oplus \mathrm{im}(\delta^*), \qquad \mathcal{H}^1 \perp \mathrm{im}(\delta) \perp \mathrm{im}(\delta^*).$$

The binding predicate is a coboundary-collision condition: if two openings (m, r) and (m', r'), over the same topology, determine the same committed cochain, then

$$\delta\big((m - m') \| (r - r')\big) = 0,$$

so a nontrivial double opening produces a nonzero element in the relevant kernel/collision relation for δ. The hiding predicate says that the randomized C^1-level value does not determine the message component. DeterministicBasis says that the blinding harmonic direction is fixed by the topology hash rather than by trusted setup. PositionBinding says that vertex positions are fixed by the topology hash, and KeyBinding says that MAC tags are bound to the chosen Hodge–Laplacian key. Full adversarial games for hiding, unforgeability, and partial openings belong to the subsequent protocol paper.

5.5 Commitment And Artifact Invariants

The commitment-facing invariants attach to the commitment payload and to the public artifact stack, not to the proximity projection. Payload well-formedness is a shape condition: it asserts the supported modulus, agreement between the declared edge count and the edge-value list, and a nonempty edge list. It is not a cryptographic uniqueness theorem and it does not assert that the cochain vector is nonzero. For example, the canonical zero payload is well formed whenever the declared edge count is positive.

Topology compatibility is a separate public invariant. Two committed cochains can be aggregated only when their moduli, declared edge counts, and topology hashes agree. The weaker same-topology relation records equality of topology hashes alone. Cryptographic binding and topology-binding claims are supplied by the commitment-security axioms, not by the raw payload record.

At the artifact level, a hashed committed cochain is a public cochain payload together with digest data and a binding predicate. A proof-carrying committed cochain adds a proof system, a statement, and a verification predicate. The basic theorem exposed to this paper is that a well-formed proof-carrying committed-cochain artifact has an underlying well-formed cochain commitment payload.

Proposition 5.5 (Proof-carrying artifact exposes payload)**.** *A well-formed proof-carrying committed-cochain artifact exposes a well-formed underlying cochain commitment payload.*

5.6 Sheaf Storage And Witness Material

Storage roots, topology hashes, cochain roots, nullifiers, and proof objects are public artifacts in the stack. Full sheaf, cochain, and fibration data remain witness material unless a protocol path explicitly exposes them. The storage mechanics are protocol-level concerns; the mathematical point here is that stored public artifacts are shadows of accepted witness data, not replacements for the hidden object.

5.7 Lossy Projection

Compression creates verifier-facing public artifacts from hidden or full witness data. The compression need not be injective; it must preserve enough structure to support the relevant public statement.

A public shadow is allowed to forget hidden data. In fact, it must forget enough data that it does not select a private representative. The relevant question is not whether the public datum is injective, but which facts survive the compression relation.

5.8 Preserved And Lost Structure

The preserved structure before the finite-code layer is:

(i) public classifier or fingerprint equality for hidden representatives;

(ii) cochain commitment payload well-formedness;

(iii) topology binding for cochain commitments;

(iv) digest/root binding for compressed public shadows;

(v) proof-carrying validity of the public statement attached to an artifact.

The lost structure is the private representative itself: full sheaf/cochain witness data, hidden transport, the operative Hodge operator, private randomness, and runtime lifecycle facts. Those may be bound to public artifacts or certified by a proof-carrying statement, but they are not made public merely by the presence of a hash, root, or commitment payload.

Byte parsing, allocation bounds, canonical encoding, concrete hash functions, domain separation, storage inclusion proofs, proof verification, state updates, and ephemeral-fiber lifecycle properties are TCB obligations. They are boundary predicates, not part of the mathematical cochain commitment payload.

Lean Correspondence

The public artifact and commitment surfaces in this section correspond to `Axioms.CochainCommitmentData`, `CochainCommitmentData.WellFormed`, `CochainCommitmentData.AddCompatible`, `CochainCommitmentData.edge_values_nonempty_of_wellFormed`, `CochainCommitmentData.modulus_gt_one_of_wellFormed`, `Axioms.CochainCommitmentAxioms`, `Axioms.BlindedCommitmentAxioms`, `Axioms.VectorCommitmentAxioms`, `Axioms.CohomologicalMACAxioms`, `TCB.HashedCommittedCochain`, `TCB.ProofCarryingCommittedCochain`, and `TCB.proofCarrying_cochain_wellFormed`.

6 Reed–Solomon Projection And MCA

The concrete projection chain belongs here because its endpoint is the Reed–Solomon received-word surface and the mutual-correlated-agreement theorem. This section uses Grassmannian data in a different role from the HCP Grassmannian of Sections 2–4. Earlier, $W \in \mathrm{Gr}(k, R^n)$ parameterizes hidden harmonic kernels and projection Laplacians $\Delta_W = I - P_W$. Here, after the hidden/geometric object has already been accepted, a frame for the Grassmannian point is used as the source of a lossy proximity projection through Plucker coordinates and actual cochain commitments toward a finite Reed–Solomon target.

6.1 Projection Chain

The staged chain has the following shape:

FrameWitness
- → Projective Plucker shadow
- → committed cochain/local syndrome
- → plane shadow
- → finite-field received word
- → canonical Reed–Solomon target.

Each stage has local proximity gates and certificates. The orchestrating projection-chain theorem composes these local certificates without rechecking them. These are proximity-specific projection gates, not the main Homotopical Encrypt acceptance layer from Section 4. Their role is to certify the lossy path into the Reed–Solomon endpoint after the main hidden/geometric

object has already been accepted. Thus the Plucker shadow is not part of the definition of HCP and is not the source of hidden-operator hardness. It is the first public projective encoding used by the proximity proof.

The semantic roles in this chain are separated. The private witness consists of the fibration data, frame choices, paths, and openings. The shadow is the Plucker/cochain/proximity data derived from that witness. A hash or root is only a compressed public handle to canonical bytes for the shadow, not the mathematical shadow itself. A certificate or SNARK proves that the private witness produced the accepted shadow and satisfied the stage invariants.

6.2 Projection-Stage Gates

The source and Plucker stages carry metric and public-shadow invariants. We use the usual Grassmannian and Plucker-coordinate algebra: a frame $(w_1, \dots, w_r)$ determines the exterior coordinate $[w_1 \wedge \cdots \wedge w_r] \in \mathbb{P}(\bigwedge^r V)$, with the quadratic Plucker relations as the projective Grassmannian equations. The source disagreement score $\rho(W)$ is required to be nonnegative and invariant under the public Plucker quotient. The Plucker metric is defined only on the admissible projective chart, and its distance must be nonnegative and frame-quotient invariant. Public-shadow respect says that equality of the public Plucker shadow determines the same committed cochain, plane shadow, and received word along this projection path.

The Plucker-to-cochain stage carries the local-syndrome interface and consumes the actual cochain commitment layer of Section 5. Its target is not an abstract stand-in for a cochain; it is the existing committed payload

$$(q, n, c, \tau)$$

with supported modulus, declared edge count, edge values, and topology hash. A concrete encoder may send valid Plucker frame data to such a well-formed payload with the expected edge count, but the storage-native statement is relational: an observed or proof-carrying cochain commitment c realizes the public Plucker shadow. This is the boundary formalized by the public-binding surface: the chain/storage layer accepts the actual commitment payload, while the proximity proof supplies the statement that this payload realizes the Plucker shadow.

Syndrome adequacy then states that a well-formed committed cochain c with vanishing local syndrome $\sigma(c) = 0$ is a global-section candidate for the chosen sheaf. In the concrete bridge, the local syndrome is the commitment-sum witness; global candidates have zero commitment sum. The point is

that the proximity proof passes through real cochain commitments and their well-formedness/topology-binding interface, not through a separate synthetic cochain carrier.

The cochain-to-plane and plane-to-RS stages carry semantic-alignment and non-collapse conditions. Semantic alignment requires valid frames, the plane shadow, and the received word to agree with the extraction maps. Plane extraction records the cochain well-formedness, expected edge count, and selected-index bounds needed to form the two-coordinate shadow. The far-region conditions assert calibrated non-collapse from the Plucker source region, through the committed cochain, to the final Reed–Solomon received word.

6.3 Canonical Reed–Solomon Endpoint

The proximity endpoint is stated over the paper-style alphabet shape $C \subseteq (\mathbb{F}^s)^n$. The canonical Homotopical Encrypt evaluation is the ordinary Reed–Solomon slice at width $s = 1$, using the standard polynomial evaluation-code convention. The canonical endpoint fixes

$$\mathbb{F} = \mathbb{Z}/12289\mathbb{Z}, \qquad n = 4096, \qquad k = 1024,$$

so the rate is $1/4$. The evaluation domain is a certified smooth power domain: the chosen generator has order 12288, and cubing it gives an element of order 4096. Hence the evaluation points are

$$1, \alpha, \alpha^2, \ldots, \alpha^{4095},$$

with $\alpha^{4096} = 1$ and no earlier repetition. The canonical radius used for the below-Johnson comparison is the absolute Hamming radius

$$2047.$$

The required finite checks are the field arithmetic, domain cardinality, smoothness, evaluation-point compatibility, and absolute-radius normalization for this endpoint; the deep list-decoding and MCA estimates enter only as cited code-theoretic theorem slots.

6.4 Mutual Correlated Agreement

Definition 6.1 (MCA bad-challenge fraction)**.** *Let $C \subseteq (\mathbb{F}^s)^n$ be a block code and let $\ell(\gamma) = f_1 + \gamma f_2$ be an affine line of received words. For a radius δ, the bad challenges are those $\gamma \in \mathbb{F}$ for which $\ell(\gamma)$ is δ-close to C on a large*

coordinate set, but the pair (f_1, f_2) is not correspondingly close to the two-fold interleaved code $C^{\oplus 2}$ on that same set. The MCA error is the fraction of bad challenges in $\mathbb{F}$.

For the canonical scalar Reed–Solomon slice $s = 1$, ordinary scalar RS words are identified with words in $(\mathbb{F}^1)^n$, and the bad-challenge predicate agrees with the paper-alphabet predicate.

Theorem 6.2 (Canonical RS MCA bound)**.** *For the canonical Homotopical Encrypt Reed–Solomon endpoint at the below-Johnson radius, every projection-induced final affine line has bad-challenge fraction at most*

$$\frac{\tau}{|\mathbb{F}|},$$

where τ is the cited MCA threshold for the canonical endpoint.

The deep list-decoding and MCA estimates are external theorem inputs: the Johnson list-decoding bound, the count and fractional MCA bounds, and the paper Definition 4.3 width-one MCA bound. The internal finite part is the endpoint plumbing, the $s = 1$ equivalence, and the specialization to projection-induced lines. The project-specific list-decoding and MCA input is the cited Proximity Prize theorem boundary [ABF26].

6.5 Corollary

Corollary 6.3 (Verifier-facing RS/MCA projection)**.** *Given an accepted hidden/geometric witness, a certified Plucker-to-cochain-to-plane-to-RS projection chain, the required hard projection-gate evidence, and a canonical Johnson-radius far-separation witness, the public verifier obtains a finite Reed–Solomon received word whose canonical MCA bad-challenge fraction is bounded by the cited threshold $\tau/|\mathbb{F}|$.*

This corollary is a proximity statement. It says that accepted hidden/geometric state can be projected to a finite public RS/MCA target with calibrated non-collapse at the selected far region. It does not say that the projection is injective, and it does not turn the RS/MCA endpoint into the source of HCP hardness.

Lean Correspondence

The proximity and finite-code endpoint in this section corresponds to the following Lean declarations: `Proximity.GrassmannianPlucker`,

`Proximity.CochainBridge`, `Proximity.PluckerCochain`, `Proximity.CochainPlane`, `Proximity.ProjectionChain`, `ProjectionChain.Chain.toEvaluationProjection`, `Proximity.Gates`, `Proximity.GateEvidence`, `CanonicalRS.BridgeData`, `CanonicalRS.CitedBounds`, `CanonicalRS.MCA`, and `CanonicalRS.MCA.projectionBadChallengeFraction_le`.

7 Future Directions

7.1 Protocol Boundary

The present paper stops at the mathematical and formalization boundary. A subsequent protocol paper should turn the commitment, oracle, proof-carrying, identity, storage, and proximity surfaces into explicit adversary games and executable verification paths. The concrete protocol must specify which certificates are public evidence, which witnesses remain private, and which runtime failures cause rejection, hold, or re-evaluation.

On the identity side, the hidden-representative/public-classifier split should be developed into explicit class-membership, unlinkability, and nullifier properties. Public identity data should remain class values, hashes, proofs, or oracle judgments; the private side should retain fibration representatives, transport data, and the Hodge operator selecting the operative kernel.

On the storage side, the commitment layer can be developed as a controlled system for sheaf/cochain projections. A public store may retain topology hashes, cochain roots, compressed shadows, and proof-carrying statements, while full sheaf and fibration witnesses remain private or are opened only through dedicated proof paths.

The protocol-facing reference point is the higher categorical cryptoeconomics program [SKdS25]. It is relevant to the design of protocol evidence and state semantics, but it is not used in the proofs of the present paper.

7.2 Complexity-Theoretic Refinements

The current formalization isolates the hidden-operator obstruction: public data does not canonically determine the operative Hodge–Laplacian kernel. It also separates the external known-lattice CVP assumption from the project-native hidden-operator ambiguity, gives public-data-only oracle insufficiency statements, records classical and quantum adversary likelihood surfaces, and proves infinitude results for the relevant Grassmannian/Laplacian families. Future work should sharpen these surfaces quantitatively: rates of growth for hidden-kernel families, refined leakage bounds for public classifiers, success

probabilities for specified adversary classes, and reductions between variants of HCP.

A Lean/Paper Correspondence

A.1 Definitions And Interfaces

HCP public observation and hidden witness.
Sections 2–3; `Hardness . PublicProjection`, `Hardness . HiddenOperatorFiber`, and `Hardness.HiddenOperatorClass`.

Matrix harmonicity and Hodge instances.
Section 2; `Hardness.Hodge_Instance` and `Hardness.MatrixIsHarmonic`.

Kernel ambiguity and lossy public data.
Sections 2–3; `Hardness . SameKernel`, `Hardness . KernelAmbiguous`, `Hardness . PublicCompatibleHodgeAmbiguity`, and `Hardness . LossyPublicProjection`.

Public-data-only Hodge solvers.
Section 3; `Hardness . PublicHodgeInstance`, `Hardness . PublicHodgeSolver`, `Hardness . PublicHodgeSolverSound`, and `Hardness.PublicHodgeSolverExactOnHarmonic`.

Explicit-lattice CVP comparison.
Section 3; `CVP . CVP _ Instance`, `CVP . CVPSolution`, and `CVP . CVPOracleCorrect`.

Hodge/CVP solution predicates.
Section 3; `Hardness . HodgeCVPSolution` and `Hardness . ProjectionHodgeCVPSolution`.

Hodge projection model.
Sections 2–3; `Laplacian.harmonicProjection`, `Laplacian.residual`, and `Laplacian.IsHarmonic`.

Categorical fibration copy surface.
Section 4; `Hardness . IsCartesianMorphism`, `Hardness . GrothendieckFibration`, `Hardness . HomotopyEquivalent`, and `Hardness.FibrationCopy`.

Semantic/public equality surface.
Section 4; `Axioms . HelferFOL`, `Axioms . HomotopyClass`, and `Axioms . Fibration`.

Heyting/oracle public judgment.
Section 4; `Axioms.Heyting3`, `Axioms.OracleQuery`, `Axioms.OracleAnswer`, and `Axioms.OracleEta2`.

Cochain commitment payload.
Section 5; `Axioms.CochainCommitmentData`, `CochainCommitmentData.WellFormed`, and `CochainCommitmentData.AddCompatible`.

Commitment security interface.
Section 5; `Axioms.CochainCommitmentAxioms`.

Proof-carrying public artifact.
Section 5; `TCB.HashedCommittedCochain` and `TCB.ProofCarryingCommittedCochain`.

Projection chain.
Section 6; `ProjectionChain.Chain` and `ProjectionChain.Chain.toEvaluationProjection`.

Proximity-stage certificates.
Section 6; `Proximity.Gates.SourceMetricGate`, `Proximity.Gates.SyndromeAdequacyGate`, and `Proximity.Gates.HardGateCertificate`.

Canonical RS/MCA endpoint.
Section 6; `CanonicalRS.BridgeData`, `CanonicalRS.MCA.Line`, and `CanonicalRS.MCA.badChallengeFraction`.

A.2 Theorems And Status

Kernel ambiguity implies lossy public projection.
`Hardness.kernelAmbiguous_implies_lossy`; Lean-proved.

Public-compatible Hodge ambiguity gives kernel ambiguity.
`Hardness.kernelAmbiguous_of_publicCompatibleHodgeAmbiguity`; Lean-proved.

Concrete Hodge ambiguity exists in nontrivial dimension.
`Hardness.Hodge_Ambiguity`, `Hardness.constantPublicProjection_publicCompatibleHodgeAmbiguity`, `Hardness.constantPublicProjection_kernelAmbiguous`, and `Hardness.constantPublicProjection_lossy`; Lean-proved.

Every explicit lattice image can be a Hodge kernel.
`Hardness.exists_matrix_ker_eq_range`; Lean-proved linear-algebra construction.

Kernel realization for explicit lattices.
`Hardness.Hodge_Hardness_Reduction` and `Hardness.Hodge_Hardness_Reduction_Proof`; Lean-proved surface.

Hodge decomposition into harmonic, exact, and coexact parts.
`Hardness.Hodge_Decomposition_Theorem`; Lean-proved theorem surface.

Ambiguous input contradicts public-data-only Hodge solving.
`Hardness.public_solver_eq_input_of_harmonic` and `Hardness.no_public_solver_on_ambiguous_input`; Lean-proved lemmas.

Known-lattice CVP does not select the hidden operator.
`Hardness.CVP_Oracle_Insufficient_For_Public_Hodge_Solving`; Lean-proved surface.

Strict hidden-operator separation.
`Hardness.Hodge_Strictly_Harder`; Lean-proved model theorem.

Projection-model Hodge recovery is CVP-shaped.
`Hardness.matrix_harmonic_iff_projection_isHarmonic`, `Hardness.Hodge_Projection_Is_CVP`, and `Hardness.harmonicProjection_Is_ProjectionHodgeCVPSolution`; Lean-proved.

Fibration copies preserve the hidden semantic surface.
`Hardness.copy_equivalent_thm` and `Hardness.FibrationCopy_is_Grothendieck`; Lean-proved categorical interface.

Exact sheaf gates supply $d^2 = 0$ and Laplacian-from-coboundary.
`Gates.structural_preconditions_of_sheaf_exact_gates`; Lean-proved alias/interface.

Well-formed payloads have nonempty edge data and supported modulus.
`CochainCommitmentData.edge_values_nonempty_of_wellFormed` and `CochainCommitmentData.modulus_gt_one_of_wellFormed`; Lean-proved.

Proof-carrying artifacts expose the committed payload.
`TCB.proofCarrying_cochain_wellFormed`; TCB boundary theorem.

Projection chain composes to one evaluation projection.
`ProjectionChain.Chain.toEvaluationProjection`; Lean-proved plumbing.

Syndrome adequacy turns zero local syndrome into codeword/global candidate.
`Proximity.Gates.SyndromeAdequacyGate.codeword_of_wellFormed_zeroSyndrome`; stage-local interface.

Final proximity evidence yields calibrated end-to-end readiness.
`Proximity.Gates.CalibratedFinalCertificate.toCalibratedEndToEndReady`; Lean-proved plumbing.

Canonical RS list/MCA bounds enter at a cited boundary.
`CanonicalRS.CitedBounds.bounds`; cited external theorem package.

Canonical MCA bad-challenge fraction is bounded after projection.
`CanonicalRS.MCA.projectionBadChallengeFraction_le`; Lean-proved from cited bounds.

B Complete Lean Coverage Ledger

This ledger is the audit trail for the translation. The body of the paper gives the mathematical definitions and proofs; this appendix records where each Lean declaration family enters the written account. Import-only aggregate files are listed with the files they collect.

B.1 Build And Aggregate Files

`lakefile.lean`.
Build configuration; no mathematical declaration is translated as a theorem.

`Axioms.lean`, `Gates.lean`, `TCB.lean`, `Consensus.lean`, `Proximity.lean`, and `Proximity/CanonicalRS.lean`.
Aggregate import surfaces. Their mathematical content is exactly the union of the component files listed below.

B.2 HCP, CVP, And Fibrational Algebra

`Hardness.lean`.
Sections 2–4 translate every declaration into the hidden Hodge-operator algebra. The HCP data are `Hodge_Instance`, `MatrixIsHarmonic`, `PublicProjection`, `PubliclyEquivalent`, `HiddenOperatorFiber`, and `HiddenOperatorClass`. The ambiguity predicates are `LossyPublicProjection`, `SameKernel`, `KernelAmbiguous`, and `PublicCompatibleHodgeAmbiguity`. The elementary closure lemmas `mem_hiddenOperatorClass_self`, `sameKernel_refl`, `kernelAmbiguous_implies_lossy`, and `kernelAmbiguous_of_publicCompatibleHodgeAmbiguity` are Theorem 2.10. The concrete ambiguity theorems `Hodge_Ambiguity`, `Hodge_Ambiguity_Proof`, `constantPublicProjection_publicCompatibleHodgeAmbiguity`, `constantPublicProjection_kernelAmbiguous`, and `constantPublicProjection_lossy` are Theorem 3.4. The known-lattice comparison declarations `Hodge_Solver_Correct`, `exists_matrix_ker_eq_range`, `Hodge_Hardness_Reduction`, and `Hodge_Hardness_Reduction_Proof` are Theorem 3.1. The decomposition theorem `Hodge_Decomposition_Theorem` is Theorem 2.5. The public-solver predicates `PublicHodgeInstance`, `PublicHodgeSolver`, `PublicHodgeSolverSound`, `PublicHodgeSolverExactOnHarmonic`, `public_solver_eq_input_of_harmonic`, `no_public_solver_on_ambiguous_input`, `Hodge_Strictly_Harder`, and `CVP_Oracle_Insufficient_For_Public_Hodge_Solving` are Theorem 3.4. The projection/CVP predicates `HodgeCVPSolution`, `ProjectionHodgeCVPSolution`, `matrix_harmonic_iff_projection_isHarmonic`, `Hodge_Projection_Is_CVP`, and `harmonicProjection_Is_ProjectionHodgeCVPSolution` are Theorem 3.2. The categorical declarations `IsCartesianMorphism`,

GrothendieckFibration, HomotopyEquivalent, Wrapper, Fibration, FibrationCopy, ToWrapper, FromWrapper, copy_equivalent_thm, and FibrationCopy_is_Grothendieck are Theorem 4.1.

CVP.lean.
Section 3 translates ExplicitLattice, CVP_Instance, InExplicitLattice, IsClosestLatticeVector, CVPSolution, CVP_Oracle, LatticeOracleCorrect, CVPOracleCorrect, CVPSolverCorrect, cvpSolverCorrect_iff_oracleCorrect, CVPHardnessAssumption, and zero_lattice_oracle_correct as the known-lattice comparison surface.

Axioms / HelferFOL . lean, Axioms / Fibration . lean, and Axioms/HomotopyClass.lean.
Section 4 translates path equality, public classifiers, public fibers, class-membership witnesses, transport, fiber systems, ephemeral fiber specifications, and nonfaithful classifiers. The equivalence, symmetry, transitivity, membership, and nonfaithfulness lemmas justify the public fiber language used in HCP.

B.3 Foundational, Sheaf, Spectral, And Judgment Axioms

Axioms/Foundation.lean and Gates/Exact.lean.
Section 4 translates Foundation and FoundationWellFormed: finite cochain data, coboundary compatibility, Hodge Laplacian formation, topology connectivity, spectral gap positivity, and public key-image/freshness premises. The exact-gate aliases ExactGateBundle, SheafExactGateBundle, FoundationWithSheafExactGates, and structural_preconditions_of_sheaf_exact_gates are Theorem 4.5.

Axioms/SheafCoherence.lean.
Section 4 translates restriction-shape compatibility, edge naturality, triangle-face commutativity, sheaf cochain complexes, Hodge coherence, finite simplicial nerves, SheafCoherence, SheafCoherenceWellFormed, FoundationWithSheafCoherence, and coherence_implies_structural_preconditions.

Axioms/Grassmannian.lean.
Sections 4 and 6 translate the Grassmannian metric and geometry objects: GrassmannianDistance, chordal_leq_geodesic, KarcherMean, SheafLaplacian, CechNerve, PluckerEligibility, ConsensusGeometry, and ConsensusGeometryWellFormed.

Axioms/SpectralBounds.lean and Gates/Spectral.lean.
Section 4 translates the constants $\sqrt{2}/2$ and $2\sqrt{2}$, the predicates SpectralGapInPoVBounds, NearPoVSpectralBoundary, TsirelsonPredicate, NearTsirelsonBoundary, the aliases SpectralGapGate and NearSpectralBoundary, the geometry witness

ByConstructionPoVGeometry, and the theorems byConstruction_gap_in_bounds and byConstruction_passes_spectral_gate.

Tsirelson.lean.
Section 4 translates the Grassmannian/Laplacian family Grassmannian_Laplacians, Ĩľ_G, LaplacianSpace, SpectralGap, Ĩľ_1, projector_map, Classical_Strategy, Quantum_Strategy, and MaxEigenvalue; the infinitude and injectivity results R_infinite, infinite_lines, Grassmannian_Laplacian_injective, projector_injective, laplacian_injective, Ĩľ_1_infinite, R_uncountable, Ĩľ_1_infinite_v2, and Ĩľ_1_Nonempty; the Laplacian-space inclusions laplacian_properties, laplacian_is_hermitian, Ĩľ_1_subset_LaplacianSpace, Ĩľ_1_subset_LaplacianSpace_v3, Ĩľ_1_subset_LaplacianSpace_v5, Ĩľ_1_subset_LaplacianSpace_explicit_v2, and Ĩľ_1_subset_LaplacianSpace_explicit; the classical/quantum statement forms Classical_Bound_Statement and Quantum_Bound_Statement; and the empty-family sanity checks Classical_Bound_empty_family and Quantum_Bound_empty_family.

Axioms/Heyting.lean and Axioms/OracleModality.lean.
Section 4 translates the three-valued Heyting algebra, meet, join, implication, negation, failure of double-negation elimination, abstract Heyting collapse, nonfaithful public truth, spectral judgments, consensus actions, oracle queries and answers, Ω_{nv}, witness collapses, quotient equivalence, sheafification, oracle forcing, η_1/η_2 oracle modalities, Lawvere–Tierney idempotence, boundary judgment behavior, canonical consensus admissibility, oracle pipelines, and the oracle/consensus bridge theorems.

Axioms / StageAdmissibility . lean, Gates / Stage . lean, and Gates/Oracle.lean.
This appendix records the stage and oracle aliases: pipeline stages, stage admissibility, transaction-critical invariant bundles, Python-stage snapshots, refinement predicates, and the theorem that a refined transaction-critical pass is admissible.

B.4 Laplacian Geometry And Search Spaces

Laplacian/Hodge.lean and Laplacian.lean.
Sections 2–3 translate harmonic projection, residual vector, residual norm, harmonicity, projection membership, nonnegativity, and all equivalences between zero residual, membership in the harmonic subspace, and harmonicity.

Laplacian/Projection.lean.
Section 3 translates the projection Laplacian $\Delta_W = I - P_W$ and its pointwise application formula.

Laplacian/Spectral.lean.
Section 4 translates sheaf Laplacians, spectral fingerprints, spectral equivalence, scalar Laplacians, kernel and eigenvalue lemmas, scalar fingerprints,

homotopy-equivalent Laplacians, and the theorem that spectrum is not homotopy invariant.

`Laplacian/Likelihood.lean`.
Section 4 records the classical and quantum adversary shapes, valid quantum adversaries, Gaussian likelihoods, minimum spectral gaps, normalized success probabilities, optimal-adversary likelihood, and the Gaussian factorization lemmas as spectral-statistical surfaces rather than HCP hardness assumptions.

`Laplacian/Grassmannian.lean`, `Laplacian/Grassmannian/Injectivity.lean`, `Laplacian / Grassmannian / SearchSpace . lean`, and `Laplacian/Grassmannian/ParametricFamily.lean`.
Sections 4 and 6 translate harmonic subspaces, the map to projection Laplacians, the families Ω_r and Ω_1, membership in Laplacian space, nonemptiness of Ω_k, injectivity of the Grassmannian Laplacian map, and the parametric construction of families of k-planes.

B.5 Commitments, Public Artifacts, And TCB Boundary

`Axioms/CochainCommitment/Payload.lean`, `Axioms/CochainCommitment/Security.lean`, and `Axioms/CochainCommitment.lean`.
Section 5 translates the canonical modulus, byte lengths, wire length, `CochainCommitmentData`, edge-count matching, nonempty edge data, supported modulus, well-formedness, commitment sums, common-topology and additive compatibility, zero and addition operations, payload lemmas, binding/hiding/additivity/security axioms, blinded and vector commitment axioms, cohomological MAC axioms, and the canonical commitment construction from foundational data.

`TCB / ProofCarrying . lean`, `TCB / Runtime . lean`, `TCB / SNARK . lean`, `TCB / CompressionBoundary . lean`, `TCB / ObservationBoundary . lean`, and `TCB/WitnessVerifier.lean`.
Section 5 and this appendix record hash words, hashed committed cochains, proof-carrying cochains, runtime specification boundaries, runtime acceptance predicates, SNARK-carrying cochains, compression contracts, compressed shadows, observation boundaries, gap-decision certificates, witness-compression envelopes, public state updates, and theorems stating that accepted artifacts expose well-formed cochain and oracle predicates.

`Axioms/ZKP.lean`.
Section 5 records the zero-knowledge proof object and its well-formedness predicate as an assumed proof-carrying artifact interface.

B.6 Projection Chain, Proximity Gates, And RS/MCA Endpoint

`Proximity / Problem . lean`, `Proximity / Public . lean`, `Proximity/Projection.lean`, and `Proximity/PluckerShadow.lean`.

Section 6 records the problem-facing public projection vocabulary that prepares the Plucker and finite-code shadow.

Proximity/Grassmannian.lean, Proximity/GrassmannianPlucker.lean, and Proximity/GrassmannianPlucker/Concrete.lean.
Section 6 translates harmonic Grassmannian points, exterior coordinates, frame witnesses, decomposability, framed systems, raw and public shadows, soundness, projective equivalence, projective shadows, projective normalization, projective framed systems, and projective sound inputs.

Proximity / CochainBridge . lean, Proximity / PluckerCochain . lean, Proximity / PluckerCochain / Concrete . lean, and Proximity/PluckerCochain/PublicBinding.lean.
Section 6 translates valid frames, Plucker-to-cochain instances, local syndrome targets, codeword predicates, evaluation projections, invariant checklists, payload encoders, commitment-sum residuals, payload realizations, hashed Plucker shadows, observed frame bindings, and the theorems that valid public shadows bind to well-formed cochains and local codewords.

Proximity / CochainPlane . lean, Proximity / CochainPlane / Core . lean, Proximity/CochainPlane/Concrete.lean, Proximity/CochainPlane/ReedSolomon.lean, and Proximity/CochainReedSolomon.lean.
Section 6 translates plane shadows, plane extraction contexts, local non-degeneracy, alignment, projected planes, plane targets, plane-to-evaluation projections, local-codeword-to-target-codeword theorems, distance monotonicity, canonical-field plane encodings, interpretations, and RS received words.

Proximity/ProjectionChain.lean and Proximity/ProjectionChain/Core.lean.
Section 6 translates the complete projection chain, including accessors for committed cochain, plane shadow, received word, and the composite evaluation projection.

Proximity / Gates . lean, Proximity / Gates / Stage . lean, Proximity/Gates/Final.lean, and Proximity/GateEvidence.lean.
Section 6 translates only proximity gates: source metric, Plucker metric, semantic alignment, syndrome adequacy, public-shadow respect, noncollapse, far-region separation, runtime binding, structural bundles, hard-gate certificates, calibrated final certificates, certified state buses, stage checklists, stage certificates, radius calibrations, final readiness, evidence slots, evidence databases, and the binding/completeness lemmas for proximity evidence.

Proximity / RS / ListDecoding . lean, Proximity / RS / ReedSolomon . lean, Proximity/RS/CorrelatedAgreement.lean, Proximity/RS/Target.lean, and Proximity/RS.lean.
Section 6 translates words, block codes, Hamming and relative Hamming balls, list-size bounds, interleaved and additive codes, RS parameters, vector codewords, correlated-agreement and mutual-correlated-agreement predicates,

bad-parameter sets, bridge obligations, calibrated *HomotopicalEncrypt*-to-RS bridges, and pullback-correctness theorems.

`Proximity/CanonicalRS/Parameters.lean`, `Proximity/CanonicalRS/PowerDomain.lean`, `Proximity/CanonicalRS/Encoding.lean`, `Proximity/CanonicalRS/Metric.lean`, `Proximity/CanonicalRS/Target.lean`, `Proximity/CanonicalRS/FarSeparation.lean`, `Proximity/CanonicalRS/BridgeData.lean`, `Proximity/CanonicalRS/BridgeWitness.lean`, `Proximity/CanonicalRS/BridgeCertificate.lean`, `Proximity/CanonicalRS/CitedBounds.lean`, `Proximity/CanonicalRS/MCA.lean`, `Proximity/CanonicalRS/MCA/Line.lean`, and `Proximity/CanonicalRS/MCA/Concrete.lean`.
Section 6 translates the canonical finite field, length, degree bound, rate, Johnson and capacity radii, smooth domains, power-root domain, canonical encoding, Hamming-to-code metrics, target bounds, far-separation witnesses, bridge data and theorem slots, bridge certificates, cited BCHKS-style bounds, MCA block code, challenge and bad-challenge fractions, and the bound on the projected bad-challenge fraction.

B.7 Subsequent Protocol And Implementation Boundary

`Axioms/Consensus.lean`, `Axioms/Ledger.lean`, `Axioms/Coregulator.lean`, and `Axioms/PythonHomotopyRefinement.lean`.
This appendix records consensus, ledger, coregulator, and Python homotopy refinement as boundary objects. Their full protocol semantics belong to the subsequent protocol paper, but their predicates are named here so the Paper 1 mathematical objects have explicit downstream interfaces.

`Consensus/Geometry.lean`, `Consensus/Reliability.lean`, and `Consensus/ProofOfValue.lean`.
This appendix records Plucker eligibility and consensus geometry aliases, reliability evidence and bounded reliability scores, proximity quality, Proof-of-Value gates, Proof-of-Value inputs, scores with and without reliability/proximity, and the score bounds. These are protocol-facing uses of the Section 4 and Section 6 mathematical predicates.

`Chain.lean`.
This appendix records validators, transactions, blocks, protocol states, spectral bounds, topology and freshness predicates, block validity, tips, heights, extension, transaction and block acceptance, ledger-state updates, key-image preservation theorems, validator eligibility, cumulative gap, preference, transaction composition and identities, quadratic weight, Sybil-resistance weighting, chain-growth monotonicity, fork-resolution transitivity, and connectedness of eligible validators. The detailed protocol architecture remains in the subsequent protocol paper.

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